

Vineth Joseph Choudary Mallavarapu

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EDUCATION

SRM Institute of Science and Technology

B-Tech in Mechatronics with specialization in Robotics | CGPA:8.16

Chennai,Tamil-Nadu

Graduation Date: May 2024

PATENTS AND PUBLICATIONS

Patents: Title: A system for identifying and weighing loose items, including Millets

Patent Application Number: 202541006107, **Publication Date:** 31/01/2025

Publication: Madhumitha G, Vineth Joseph Mallavarapu, and Leela Ram Chelli, 2024, "**Development of Automated Millet Identification and Billing System,**" ADCIS, 3.

Certifications: ROS, IoT, Solidworks.

EXPERIENCE

Grace Drone Technologies

Hyderabad, Telangana

Drone operation support

Oct 2024 - Sep 2025

- Worked extensively on **DJI MATRICE 30T**. Assisted in pre-flight and post-flight operations: drone calibration, battery swaps, equipment setup, and flight safety checks.
- Supported **mission planning and aerial surveys** using DJI softwares and DroneDeploy for automated flight path planning, mapping, and data collection.
- Contributed to survey and inspection projects by processing aerial data and preparing site layouts and measurements with AutoCAD.
- Managed data handling and visualization, including transferring drone imagery, generating orthomosaics, and preparing outputs for client reports.
- Collaborated with pilots and engineers to ensure safe, efficient, and regulation-compliant operations during field deployments.

Fronius India Pvt Ltd

Pune,Maharashtra

Intern

May 2024 - Jun 2024

- Acquired proficiency in welding techniques (**SMAW, GMAW, TIG**) and process control methods.
- Conducted experimental analysis on a **FANUC 6-axis robotic manipulator** to assess shielding gas effects on mild steel welds.
- Applied advanced welding modes (**CMT, PMC, PULSE, LSC, and standard**) to optimize joint quality.

Apsis Sollutions

Remote

Intern

Feb 2024 - Mar 2024

- Worked on IoT projects, including a **smart factory system**, an **ESP32-based weather station**, and a **voice-controlled home automation** platform for improved efficiency.

Vaayusastra

Chennai,Tamil Nadu

Mechanical Engineer Intern

Feb 2023 - May 2023

- Designed and fabricated an **autonomous obstacle avoidance vehicle**.
- Designed a **rocker bogie mechanism** using **SolidWorks** for conceptual and mechanical layout.

MYEQUATION

Remote

Mechanical Designer

May 2021 - Jun 2021

- Designed innovative electric cycle concepts by repurposing existing bicycle chassis for new electric models.

SKILLS & LANGUAGES

Skills: Robotic welding, SolidWorks, Ansys, AutoCAD, Arduino IDE, Raspberry Pi, MATLAB, Microsoft Office (Word, Excel, and PowerPoint), and ROS 2.

Languages: Embedded C,Python,C++

PROJECTS

Autonomous Mobile Robot Using Slam Algorithm

Hyderabad,Telangana

Feb 2025 - Jul 2025

- Developed an autonomous mobile robot using **ROS 2 Jazzy**, implementing **SLAM** for real-time mapping and navigation.
- Integrated an LLaMA-based **LLM** for **voice-controlled interaction and chat-based interaction**, enabling natural language command execution.
- Achieved seamless fusion of robotics and AI, enhancing autonomous decision-making and user experience.

Automated Millet Identification and Billing System

Chennai,Tamil Nadu

Jan 2024 - May 2024

- Built an automated millet classification system that uses **deep learning** for accurate millet variety identification via an upper camera with an **accuracy of 85%**, coupled with load cells for weight estimation. The system autonomously generates a user bill based on the data and also features an admin portal for updating the prices and checking billing history.

Automated Conveyor Sorting and Assembly System

Chennai,Tamil Nadu

Jun 2023 - Nov 2023

- Fabricated and developed a robotic workcell for efficient **colour-based object sorting and assembly**, featuring an **RRR manipulator (calculated inverse kinematics)**, conveyor, and an IoT-based control system for streamlined operations.

Probabilistic route mapping for a mobile robot

Chennai,Tamil Nadu

- Mapped an indoor arena for a mobile robot using a binary occupancy grid, simulating probabilistic route map to determine the **most efficient path between designated points** on MATLAB.

CAMPUS ENGAGEMENT

Powertrain Engineer	:	Infieon SuperMileage (a student team from the SRM Institute of Science and Technology participates in the Shell Eco Marathon competition annually).
Social Service Lead	:	Worked as a social service lead at the Association of Mechatronics at the SRM Institute of Science and Technology for two years.